Erratum

Erratum to "Automatic Detection and Identification of Floating Marine Debris Using Multispectral Satellite Imagery"

Miguel M. Duarte and Leonardo Azevedo

In the above article [1], Figs. 9, 10, and 13 were mistakenly duplicated, as were Figs. 11, 14, and 16. As a result, the reported results were inaccurate. The correct versions of these figures and their respective captions are provided here, ensuring the accurate interpretation of the findings presented in the article.

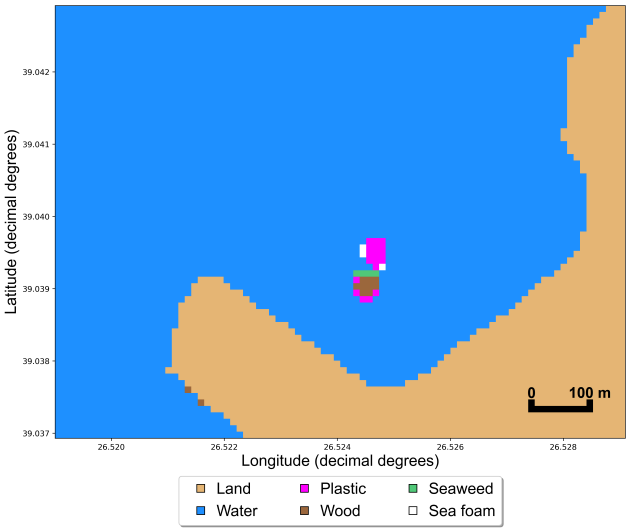


Fig. 9. Predictions from the XGBoost model that does not quantify uncertainty. The original data are from a Sentinel-2A image from July 31, 2021, in the Gulf of Gera, Greece. This corresponds to Fig. 9 in the above article [1].

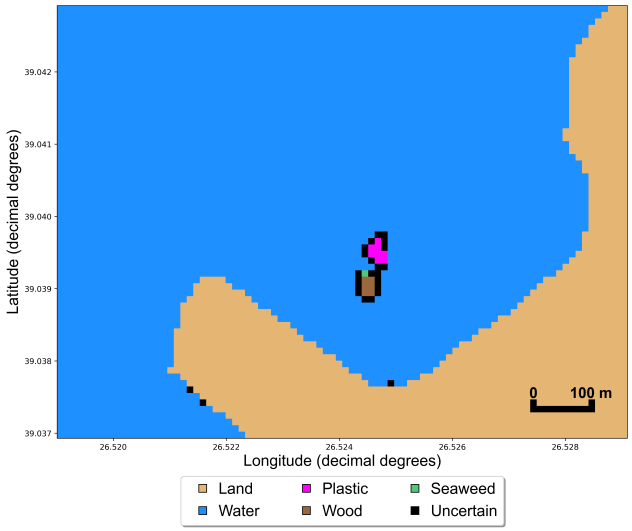


Fig. 10. Ensemble model's predictions. The original data are from a Sentinel-2A image from July 31, 2021, in the Gulf of Gera, Greece. This corresponds to Fig. 10 in the above article [1].

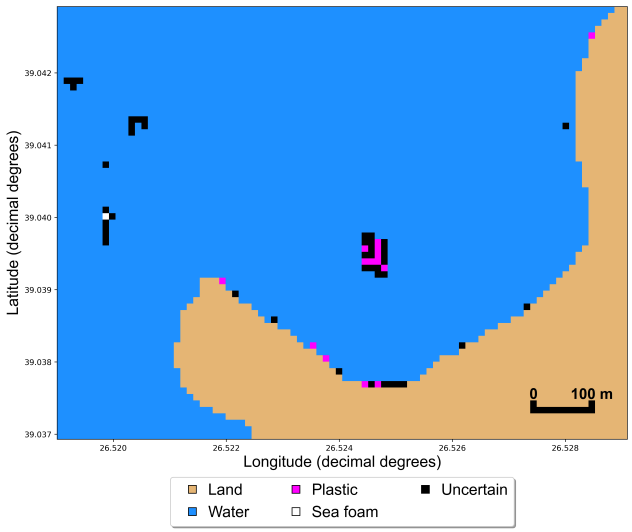


Fig. 13. Predictions from a Sentinel-2B image from September 4, 2021, in the Gulf of Gera, Greece. On this day, another experience for the PLP 2021 was performed. According to the authors, the sky was clear, but the sunlight intensity was lower than in previous experiments, and the weather was windy (8.5 m/s) [2]. They removed the wooden target and deployed some of it under the plastic target, simulating a mixed target that is closer to what is found in the ocean. The new target Fig. 13.

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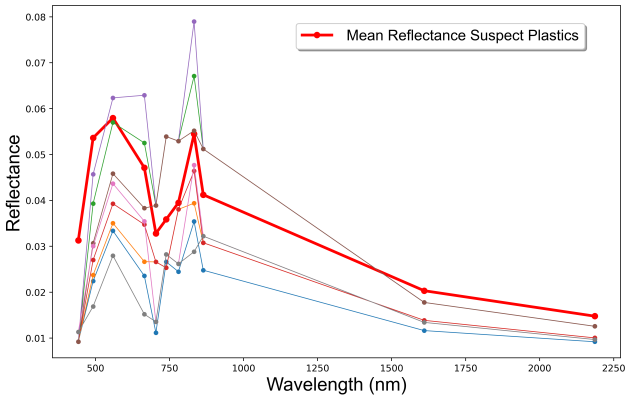


Fig. 11. Mean reflectance of all data classified as plastic used to train the model (red line) and the individual spectra of pixels identified as plastic by the model in Fig. 10 (shown in various colors). This corresponds to Fig. 11 in the above article [1].

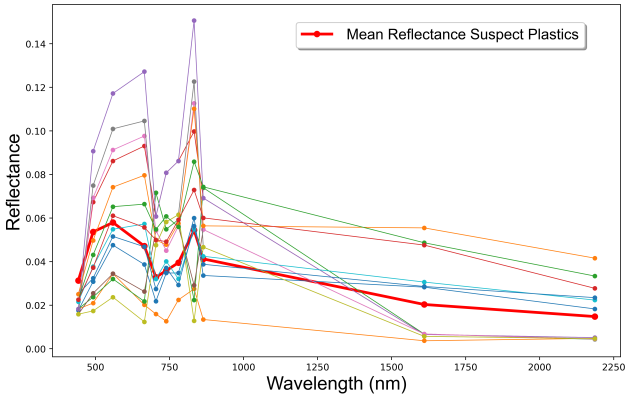


Fig. 14. Mean reflectance of all data classified as plastic used to train the model (red line) and the individual spectra of pixels identified as plastic by the model in Fig. 11 (shown in various colors). This corresponds to Fig. 14 in the above article [1].

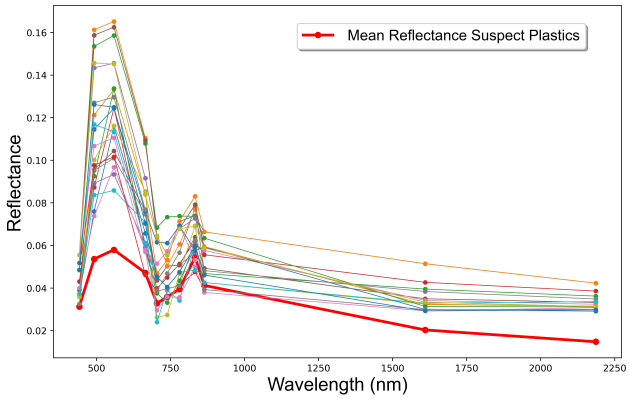


Fig. 16. Mean reflectance of all data classified as plastic used to train the model (red line) and the individual spectra of pixels identified as plastic by the model in Fig. 15 of the above article (shown in various colors). This corresponds to Fig. 16 in the above article [1].

ACKNOWLEDGMENT

The datasets used in this study are available at https://github.com/miguelmendesduarte/Floating-Marine-Debris-Data.

REFERENCES

[1] M. M. Duarte and L. Azevedo, "Automatic detection and identification of floating marine debris using multispectral satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2002315, doi: 10.1109/TGRS.2023.3283607.

[2] Marine Remote Sensing Group. *Plastic Litter Project 2021*. Accessed: Dec. 18, 2022. [Online]. Available: http://plp.aegean.gr/